

**Q1.** Applying the principle of homogeneity of dimensions, determine which one is correct, where  $T$  is time period,  $G$  is gravitational constant,  $M$  is mass,  $r$  is radius of orbit.

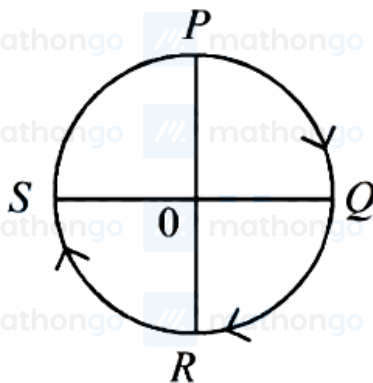
(1)  $T^2 = \frac{4\pi^2 r^2}{GM}$

(2)  $T^2 = \frac{4\pi^2 r}{GM^2}$

(3)  $T^2 = \frac{4\pi^2 r^3}{GM}$

(4)  $T^2 = 4\pi^2 r^3$

**Q2.** A cyclist starts from the point  $P$  of a circular ground of radius 2 km and travels along its circumference to the



point  $S$ . The displacement of a cyclist is:

(1)  $\sqrt{8}$  km

(2) 8 km

(3) 6 km

(4) 4 km

**Q3.** A 2 kg brick begins to slide over a surface which is inclined at an angle of  $45^\circ$  with respect to horizontal axis.

The co-efficient of static friction between their surfaces is:

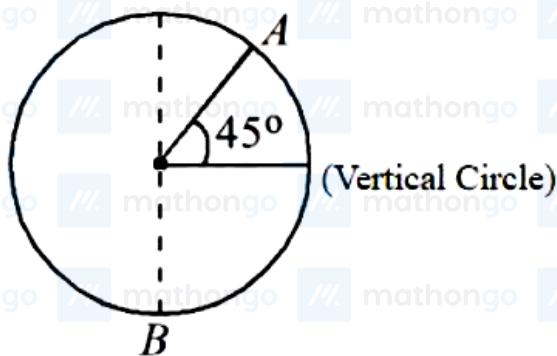
(1) 1.7

(2)  $\frac{1}{\sqrt{3}}$

(3) 0.5

(4) 1

**Q4.** A body of  $m$  kg slides from rest along the curve of vertical circle from point  $A$  to  $B$  in friction less path. The



velocity of the body at  $B$  is:

(given,  $R = 14$  m,  $g = 10$  m/s<sup>2</sup> and  $\sqrt{2} = 1.4$ )

(1) 16.7 m/s

(2) 19.8 m/s

(3) 10.6 m/s

(4) 21.9 m/s

**Q5.** A 90 kg body placed at  $2R$  distance from surface of earth experiences gravitational pull of :

(  $R$  = Radius of earth,  $g = 10$  m s<sup>-2</sup> )

(1) 100 N

(2) 300 N

(3) 225 N

(4) 120 N

**Q6.** Correct formula for height of a satellite from earths surface is:

$$(1) \left( \frac{T^2 R^2}{4\pi^2 g} \right)^{1/3} - R$$

$$(2) \left( \frac{T^2 R^2 g}{4\pi^2} \right)^{1/3} - R$$

$$(3) \left( \frac{T^2 R^2 g}{4\pi^2} \right)^{-1/3} + R$$

$$(4) \left( \frac{T^2 R^2 g}{4\pi^2} \right)^{1/2} - R$$

**Q7.** Given below are two statements : Statement I : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well. Statement II : The rise of a liquid in a capillary tube does not depend on the inner radius of the tube. In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are false.

(4) Both Statement I and Statement II are true.

**Q8.** A sample of gas at temperature  $T$  is adiabatically expanded to double its volume. Adiabatic constant for the gas is  $\gamma = 3/2$ . The work done by the gas in the process is: ( $\mu = 1$  mole)

$$(1) RT[1 - 2\sqrt{2}]$$

$$(2) RT[\sqrt{2} - 2]$$

$$(3) RT[2 - \sqrt{2}]$$

$$(4) RT[2\sqrt{2} - 1]$$

**Q9.** The translational degrees of freedom ( $f_t$ ) and rotational degrees of freedom ( $f_r$ ) of  $\text{CH}_4$  molecule are:

$$(1) f_t = 2 \text{ and } f_r = 2$$

$$(2) f_t = 3 \text{ and } f_r = 3$$

$$(3) f_t = 3 \text{ and } f_r = 2$$

$$(4) f_t = 2 \text{ and } f_r = 3$$

**Q10.** In simple harmonic motion, the total mechanical energy of given system is  $E$ . If mass of oscillating particle  $P$



is doubled then the new energy of the system for same amplitude is:

$$(1) E$$

$$(2) E/\sqrt{2}$$

$$(3) 2E$$

$$(4) E\sqrt{2}$$

**Q11.** A charge  $q$  is placed at the center of one of the surface of a cube. The flux linked with the cube is:

$$(1) \frac{q}{2\epsilon_0}$$

$$(2) \frac{q}{8\epsilon_0}$$

$$(3) \text{Zero}$$

$$(4) \frac{q}{4\epsilon_0}$$

**Q12.** An electric bulb rated 50 W – 200 V is connected across a 100 V supply. The power dissipation of the bulb is:

$$(1) 25 \text{ W}$$

$$(2) 12.5 \text{ W}$$

$$(3) 50 \text{ W}$$

$$(4) 100 \text{ W}$$

**Q13.** The magnetic moment of a bar magnet is  $0.5 \text{ Am}^2$ . It is suspended in a uniform magnetic field of  $8 \times 10^{-2} \text{ T}$ .

The work done in rotating it from its most stable to most unstable position is:

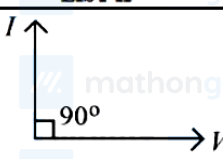
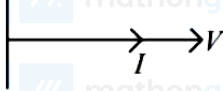
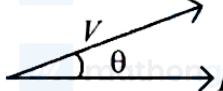
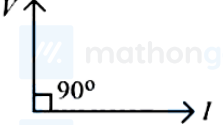
$$(1) 8 \times 10^{-2} \text{ J}$$

$$(2) 4 \times 10^{-2} \text{ J}$$

$$(3) \text{Zero}$$

$$(4) 16 \times 10^{-2} \text{ J}$$

Q14.

| LIST I |                           | LIST II |                                                                                    |
|--------|---------------------------|---------|------------------------------------------------------------------------------------|
| A.     | Purely capacitive circuit | I.      |  |
| B.     | Purely inductive circuit  | II.     |  |
| C.     | LCR series at resonance   | III.    |  |
| D.     | LCR series circuit        | IV.     |  |

Match List I with List II

Choose the correct answer from the options given below:

(1) A-I, B-IV, C-II, D-III

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-I, C-III, D-II

(4) A-I, B-IV, C-III, D-II

Q15. Arrange the following in the ascending order of wavelength: A. Gamma rays ( $\lambda_1$ ) B.  $x$  - rays ( $\lambda_2$ ) C. Infrared waves ( $\lambda_3$ ) D. Microwaves ( $\lambda_4$ ) Choose the most appropriate answer from the options given below

(1)  $\lambda_4 < \lambda_3 < \lambda_1 < \lambda_2$ (2)  $\lambda_2 < \lambda_1 < \lambda_4 < \lambda_3$ (3)  $\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$ (4)  $\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$ 

Q16. The width of one of the two slits in a Young's double slit experiment is 4 times that of the other slit. The ratio of the maximum of the minimum intensity in the interference pattern is:

(1) 1 : 1

(2) 4 : 1

(3) 9 : 1

(4) 16 : 1

Q17. Given below are two statements: one is labelled as Assertion **A** and the other is labelled as Reason **R**.

Assertion A: Number of photons increases with increase in frequency of light. Reason R: Maximum kinetic energy of emitted electrons increases with the frequency of incident radiation. In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are correct and **R** is NOT the correct explanation of **A**.

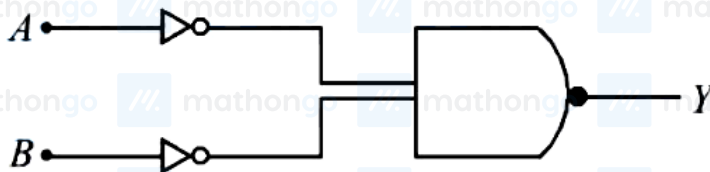
(3) **A** is not correct but **R** is correct.

(4) **A** is correct but **R** is not correct.

Q18. According to Bohr's theory, the moment of momentum of an electron revolving in 4<sup>th</sup> orbit of hydrogen atom is:

(1)  $\frac{h}{\pi}$ (2)  $\frac{h}{2\pi}$ (3)  $8\frac{h}{\pi}$ (4)  $2\frac{h}{\pi}$

Q19.

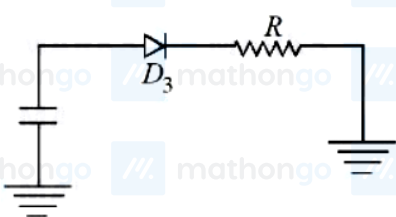


Identify the logic gate given in the circuit:

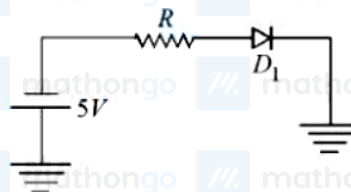
- (1) NAND-gate (2) AND gate  
(3) NOR gate (4) OR-gate

Q20. Which of the diode circuit shows correct biasing used for the measurement of dynamic resistance of p-n junction diode :

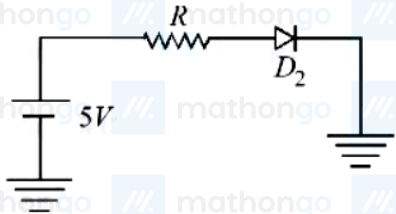
(1)



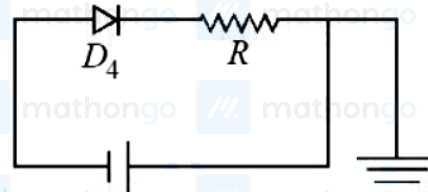
(2)



(3)



(4)



Q21. A bus moving along a straight highway with speed of 72 km/h is brought to halt within 4s after applying the brakes. The distance travelled by the bus during this time (Assume the retardation is uniform) is \_\_\_\_ m.

Q22. In a system two particles of masses  $m_1 = 3$  kg and  $m_2 = 2$  kg are placed at certain distance from each other.

The particle of mass  $m_1$  is moved towards the center of mass of the system through a distance 2 cm. In order to keep the center of mass of the system at the original position, the particle of mass  $m_2$  should move towards the center of mass by the distance \_\_\_\_ cm.

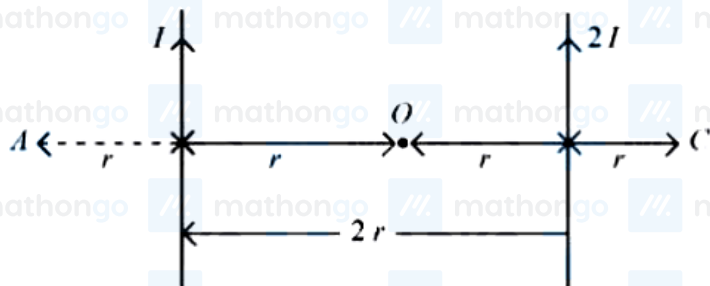
Q23. Mercury is filled in a tube of radius 2 cm up to a height of 30 cm. The force exerted by mercury on the bottom of the tube is \_\_\_\_ N. (Given, atmospheric pressure =  $10^5 \text{ Nm}^{-2}$ , density of mercury =  $1.36 \times 10^4 \text{ kg m}^{-3}$ ,  $g = 10 \text{ m s}^{-2}$ ,  $\pi = \frac{22}{7}$ )

Q24. The displacement of a particle executing SHM is given by  $x = 10 \sin \left( \omega t + \frac{\pi}{3} \right) \text{ m}$ . The time period of motion is 3.14 s. The velocity of the particle at  $t = 0$  is \_\_\_\_ m/s.

Q25. A parallel plate capacitor of capacitance 12.5pF is charged by a battery connected between its plates to potential difference of 12.0 V. The battery is now disconnected and a dielectric slab ( $\epsilon_r = 6$ ) is inserted between the plates. The change in its potential energy after inserting the dielectric slab is \_\_\_\_  $10^{-12}$  J.

Q26. Two wires A and B are made up of the same material and have the same mass. Wire A has radius of 2.0 mm and wire B has radius of 4.0 mm. The resistance of wire B is  $2\Omega$ . The resistance of wire A is \_\_\_\_  $\Omega$ .

Q27. Two parallel long current carrying wire separated by a distance  $2r$  are shown in the figure. The ratio of magnetic field at  $A$  to the magnetic field produced at  $C$  is  $\frac{x}{7}$ . The value of  $x$  is \_\_\_\_\_



Q28. A rod of length 60 cm rotates with a uniform angular velocity  $20\text{rad s}^{-1}$  about its perpendicular bisector, in a uniform magnetic field  $0.5T$ . The direction of magnetic field is parallel to the axis of rotation. The potential difference between the two ends of the rod is \_\_\_\_\_ V.

Q29. A light ray is incident on a glass slab of thickness  $4\sqrt{3}$  cm and refractive index  $\sqrt{2}$ . The angle of incidence is equal to the critical angle for the glass slab with air. The lateral displacement of ray after passing through glass slab is \_\_\_\_\_ cm. ( Given  $\sin 15^\circ = 0.25$  )

Q30. The disintegration energy  $Q$  for the nuclear fission of  $^{235}\text{U} \rightarrow ^{140}\text{Ce} + ^{94}\text{Zr} + n$  is \_\_\_\_\_ MeV. Given atomic masses of  $^{235}\text{U} : 235.0439u$ ;  $^{140}\text{Ce} : 139.9054u$ ;  $^{94}\text{Zr} : 93.9063u$ ;  $n : 1.0086u$ , Value of  $c^2 = 931\text{MeV/u}$

Q31. Choose the Incorrect Statement about Dalton's Atomic Theory

- |                                                                                 |                                                                                                                                                                      |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (1) chemical reactions involve reorganization of atoms                          | (2) Matter consists of indivisible atoms.                                                                                                                            |
| (3) Compounds are formed when atoms of different elements combine in any ratio. | (4) Compounds are formed when atoms of different elements combine in any ratio. All the atoms of a given element have identical properties including identical mass. |

Q32. The correct order of the first ionization enthalpy is

- |                                         |                                        |
|-----------------------------------------|----------------------------------------|
| (1) $\text{Al} > \text{Ga} > \text{Tl}$ | (2) $\text{Ga} > \text{Al} > \text{B}$ |
| (3) $\text{Tl} > \text{Ga} > \text{Al}$ | (4) $\text{B} > \text{Al} > \text{Ga}$ |

Q33. Given below are two statements : Statement I : The correct order of first ionization enthalpy values of Li, Na, F and Cl is  $\text{Na} < \text{Li} < \text{Cl} < \text{F}$ . Statement II : The correct order of negative electron gain enthalpy values of Li, Na, F and Cl is  $\text{Na} < \text{Li} < \text{F} < \text{Cl}$  In the light of the above statements, choose the correct answer from the options given below :

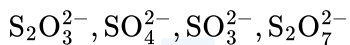
- |                                                   |                                                   |
|---------------------------------------------------|---------------------------------------------------|
| (1) Statement I is true but Statement II is false | (2) Statement I is false but Statement II is true |
| (3) Both Statement I and Statement II are false   | (4) Both Statement I and Statement II are true    |

Q34. The correct statement/s about Hydrogen bonding is/are A. Hydrogen bonding exists when H is covalently bonded to the highly electro negative atom. B. Intermolecular H bonding is present in o-nitro phenol C. Intramolecular H bonding is present in HF. D. The magnitude of H bonding depends on the physical state of

the compound. E. H-bonding has powerful effect on the structure and properties of compounds Choose the correct answer from the options given below:

- (1) A, B, D only (2) A, D, E only  
(3) A only (4) A, B, C only

**Q35.** The number of species from the following that have pyramidal geometry around the central atom is \_\_\_\_\_.

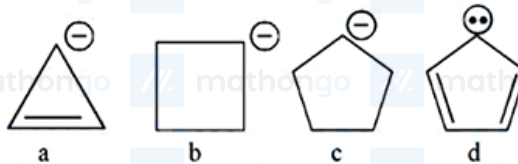


- (1) 4 (2) 3  
(3) 2 (4) 1

**Q36.** The equilibrium constant for the reaction  $\text{SO}_3(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$  is  $K_c = 4.9 \times 10^{-2}$ . The value of  $K_c$  for the reaction given below is  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$  is :

- (1) 4.9 (2) 49  
(3) 41.6 (4) 416

**Q37.**



Correct order of stability of carbanion is -

- (1)  $d > a > c > b$  (2)  $a > b > c > d$   
(3)  $d > c > b > a$  (4)  $c > b > d > a$

**Q38.** Common name of Benzene - 1, 2 - diol is -

- (1) catechol (2) o-cresol  
(3) quinol (4) resorcinol

**Q39.** The adsorbent used in adsorption chromatography is/are - A. silica gel B. alumina C. quick lime D. magnesia

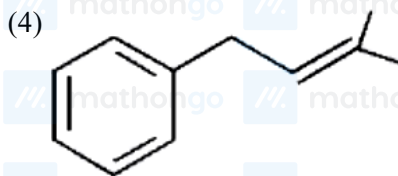
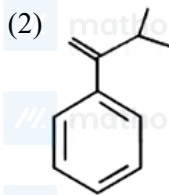
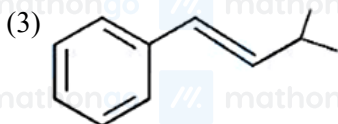
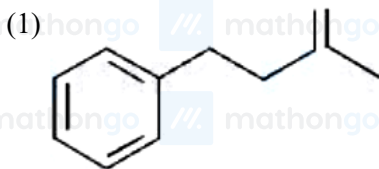
Choose the most appropriate answer from the options given below :

- (1) A only (2) B only  
(3) C and D only (4) A and B only

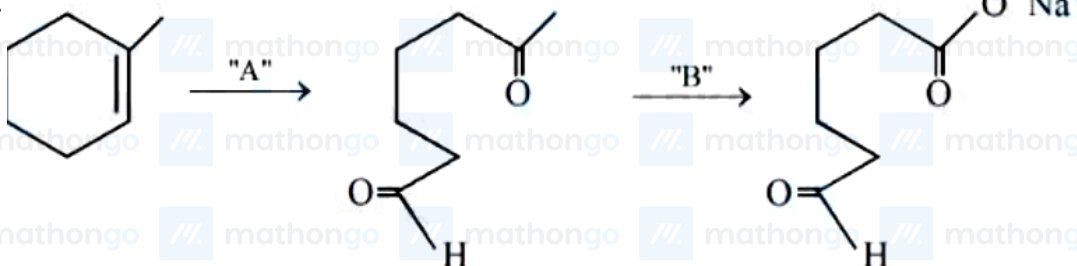
**Q40.**



Product P is



Q41.



In the above chemical reaction sequence "A" and "B" respectively are

- (1)  $\text{H}_2\text{O}$ ,  $\text{H}^+$  and  $\text{KMnO}_4$  (2)  $\text{O}_3$ ,  $\text{Zn}/\text{H}_2\text{O}$  and  $\text{NaOH}_{(\text{alc})}/\text{I}_2$   
 (3)  $\text{O}_3$ ,  $\text{Zn}/\text{H}_2\text{O}$  and  $\text{KMnO}_4$  (4)  $\text{H}_2\text{O}$ ,  $\text{H}^+$  and  $\text{NaOH}_{(\text{alc})}/\text{I}_2$

Q42. For a strong electrolyte, a plot of molar conductivity against (concentration) $^{1/2}$  is a straight line, with a negative slope, the correct unit for the slope is

- (1)  $\text{Scm}^2 \text{mol}^{-3/2} \text{L}^{-1/2}$  (2)  $\text{Scm}^2 \text{mol}^{-3/2} \text{L}^{1/2}$   
 (3)  $\text{Scm}^2 \text{mol}^{-3/2} \text{L}$  (4)  $\text{Scm}^2 \text{mol}^{-1} \text{L}^{1/2}$

Q43. Fuel cell, using hydrogen and oxygen as fuels, A. has been used in spaceship B. has an efficiency of 40% to produce electricity C. uses aluminum as catalysts D. is eco-friendly E. is actually a type of Galvanic cell only. Choose the correct answer from the options given below:

- (1) A, B, D, E only (2) A, D, E only  
 (3) A, B, D only (4) A, B, C only

Q44. When  $\text{MnO}_2$  and  $\text{H}_2\text{SO}_4$  is added to a salt (A), the greenish yellow gas liberated as salt (A) is :

- (1)  $\text{CaI}_2$  (2)  $\text{NaBr}$   
 (3)  $\text{KNO}_3$  (4)  $\text{NH}_4\text{Cl}$

Q45. A first row transition metal in its +2 oxidation state has a spin-only magnetic moment value of 3.86 BM. The atomic number of the metal is

- (1) 26 (2) 25  
 (3) 23 (4) 22

Q46. If an iron (III) complex with the formula  $[\text{Fe}(\text{NH}_3)_x(\text{CN})_y]^-$  has no electron in its  $e_g$  orbital, then the value of  $x + y$  is

(1) 4

(3) 6

(2) 5

(4) 3

Q47. The number of unpaired d-electrons in  $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$  is

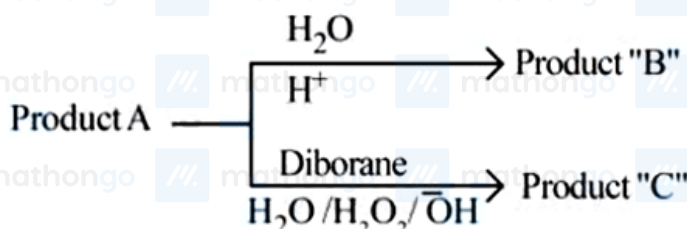
(1) 2

(3) 0

(2) 1

(4) 4

Q48.  $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br} + \text{NaOH} \xrightarrow{\text{C}_2\text{H}_5\text{OH}}$  Product 'A'



Consider the above reactions, identify product B and product C.

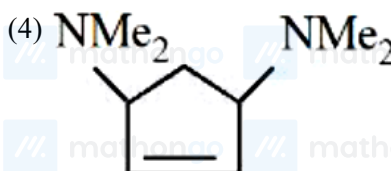
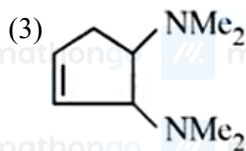
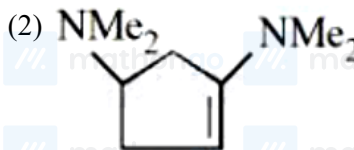
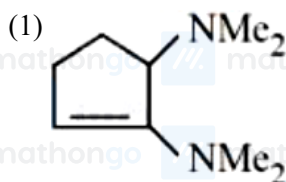
(1) B = 1-Propanol C = 2-Propanol

(2) B = C = 2-Propanol

(3) B = 2-Propanol C = 1-Propanol

(4) B = C = 1-Propanol

Q49. Find out the major product formed from the following reaction.  $[\text{Me} : -\text{CH}_3]$



Q50.

Match List I with List II

|    | List - I                                    |      | List - II          |
|----|---------------------------------------------|------|--------------------|
| A. | $\alpha$ - Glucose and $\alpha$ - Galactose | I.   | Functional isomers |
| B. | $\alpha$ - Glucose and $\beta$ - Glucose    | II.  | Homologous         |
| C. | $\alpha$ - Glucose and $\alpha$ - Fructose  | III. | Anomers            |
| D. | $\alpha$ - Glucose and $\alpha$ - Ribose    | IV.  | Epimers            |

Choose the

correct answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-III, C-II, D-I

(4) A-III, B-IV, C-II, D-I

Q51. The maximum number of orbitals which can be identified with  $n = 4$  and  $m_l = 0$  is \_\_\_\_\_

Q52. Number of compounds / species from the following with non-zero dipole moment is \_\_\_\_\_

$\text{BeCl}_2, \text{BCl}_3, \text{NF}_3, \text{XeF}_4, \text{CCl}_4, \text{H}_2\text{O}, \text{H}_2\text{S}, \text{HBr}, \text{CO}_2, \text{H}_2, \text{HCl}$

Q53. Three moles of an ideal gas are compressed isothermally from 60 L to 20 L using constant pressure of 5 atm.

Heat exchange  $Q$  for the compression is - \_\_\_\_\_ Lit. atm.

Q54. The total number of 'sigma' and 'Pi' bonds in 2-oxohex-4-ynoic acid is \_\_\_\_\_

Q55. 2.7 kg of each of water and acetic acid are mixed. The freezing point of the solution will be  $-x^\circ\text{C}$ . Consider the acetic acid does not dimerise in water, nor dissociates in water.  $x =$  \_\_\_\_\_ (nearest integer) [Given: Molar mass of water =  $18 \text{ g mol}^{-1}$ , acetic acid =  $60 \text{ g mol}^{-1}$   $K_f \text{H}_2\text{O} : 1.86 \text{ K kg mol}^{-1}$   $K_f$  acetic acid:  $3.90 \text{ K kg mol}^{-1}$  freezing point:  $\text{H}_2\text{O} = 273 \text{ K}$ , acetic acid =  $290 \text{ K}$ ]

Q56. Consider the following reaction, the rate expression of which is given below



$$\text{rate} = k[\text{A}]^{1/2}[\text{B}]^{1/2}$$

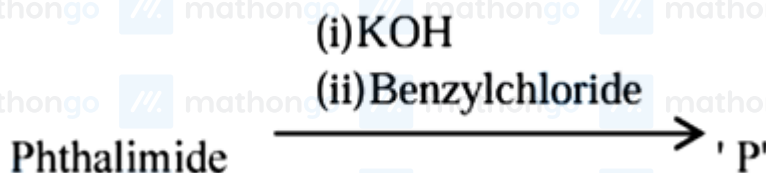
The reaction is initiated by taking 1M concentration of A and B each. If the rate constant ( $k$ ) is  $4.6 \times 10^{-2} \text{ s}^{-1}$ , then the time taken for A to become 0.1M is \_\_\_\_\_ sec. (nearest integer)

Q57. A first row transition metal with highest enthalpy of atomisation, upon reaction with oxygen at high

temperature forms oxides of formula  $\text{M}_2\text{O}_n$  (where  $n = 3, 4, 5$ ). The 'spin-only' magnetic moment value of the amphoteric oxide from the above oxides is \_\_\_\_\_ BM (near integer) (Given atomic number:

Sc : 21, Ti : 22, V : 23, Cr : 24, Mn : 25, Fe : 26, Co : 27, Ni : 28, Cu : 29, Zn : 30)

Q58. Phthalimide is made to undergo following sequence of reactions.



Total number of  $\pi$  bonds present in product 'P' is/are \_\_\_\_\_

Q59. From 6.55 g of aniline, the maximum amount of acetanilide that can be prepared will be \_\_\_\_\_  $\times 10^{-1}$  g.

Q60. Vanillin compound obtained from vanilla beans, has total sum of oxygen atoms and  $\pi$  electrons is \_\_\_\_\_

Q61. The area (in sq. units) of the region  $S = \{z \in \mathbb{C} : |z - 1| \leq 2; (z + \bar{z}) + i(z - \bar{z}) \leq 2, \text{Im}(z) \geq 0\}$  is

- (1)  $\frac{7\pi}{3}$   
(3)  $\frac{17\pi}{8}$

- (2)  $\frac{7\pi}{4}$   
(4)  $\frac{3\pi}{2}$

Q62. The value of  $\frac{1 \times 2^2 + 2 \times 3^2 + \dots + 100 \times (101)^2}{1^2 \times 2 + 2^2 \times 3 + \dots + 100^2 \times 101}$  is

- (1)  $\frac{32}{31}$   
(3)  $\frac{306}{305}$

- (2)  $\frac{31}{30}$   
(4)  $\frac{305}{301}$

**Q63.** Let three real numbers  $a, b, c$  be in arithmetic progression and  $a + 1, b, c + 3$  be in geometric progression. If  $a > 10$  and the arithmetic mean of  $a, b$  and  $c$  is 8, then the cube of the geometric mean of  $a, b$  and  $c$  is

- (1) 128 (2) 316  
(3) 120 (4) 312

**Q64.** If the coefficients of  $x^4, x^5$  and  $x^6$  in the expansion of  $(1 + x)^n$  are in the arithmetic progression, then the maximum value of  $n$  is:

- (1) 7 (2) 21  
(3) 28 (4) 14

**Q65.** Let  $C$  be a circle with radius  $\sqrt{10}$  units and centre at the origin. Let the line  $x + y = 2$  intersects the circle  $C$  at the points  $P$  and  $Q$ . Let  $MN$  be a chord of  $C$  of length 2 unit and slope  $-1$ . Then, a distance (in units) between the chord  $PQ$  and the chord  $MN$  is

- (1)  $3 - \sqrt{2}$  (2)  $\sqrt{2} + 1$   
(3)  $\sqrt{2} - 1$  (4)  $2 - \sqrt{3}$

**Q66.** Let  $PQ$  be a chord of the parabola  $y^2 = 12x$  and the midpoint of  $PQ$  be at  $(4, 1)$ . Then, which of the following point lies on the line passing through the points  $P$  and  $Q$ ?

- (1)  $(3, -3)$  (2)  $(2, -9)$   
(3)  $(\frac{3}{2}, -16)$  (4)  $(\frac{1}{2}, -20)$

**Q67.** Consider a hyperbola  $H$  having centre at the origin and foci on the  $x$ -axis. Let  $C_1$  be the circle touching the hyperbola  $H$  and having the centre at the origin. Let  $C_2$  be the circle touching the hyperbola  $H$  at its vertex and having the centre at one of its foci. If areas (in sq units) of  $C_1$  and  $C_2$  are  $36\pi$  and  $4\pi$ , respectively, then the length (in units) of latus rectum of  $H$  is

- (1)  $\frac{14}{3}$  (2)  $\frac{28}{3}$   
(3)  $\frac{11}{3}$  (4)  $\frac{10}{3}$

**Q68.** Let  $f(x) = \int_0^x (t + \sin(1 - e^t)) dt, x \in \mathbb{R}$ . Then,  $\lim_{x \rightarrow 0} \frac{f(x)}{x^3}$  is equal to

- (1)  $-\frac{1}{6}$  (2)  $\frac{2}{3}$   
(3)  $-\frac{2}{3}$  (4)  $\frac{1}{6}$

**Q69.** If the mean of the following probability distribution of a random variable  $X$ :

|        |     |      |         |      |      |
|--------|-----|------|---------|------|------|
| $X$    | 0   | 2    | 4       | 6    | 8    |
| $P(X)$ | $a$ | $2a$ | $a + b$ | $2b$ | $3b$ |

is  $\frac{46}{9}$ , then the variance of the distribution is

- (1)  $\frac{173}{27}$  (2)  $\frac{566}{81}$   
(3)  $\frac{151}{27}$  (4)  $\frac{581}{81}$

**Q70.** Let a relation  $R$  on  $N \times N$  be defined as:  $(x_1, y_1)R(x_2, y_2)$  if and only if  $x_1 \leq x_2$  or  $y_1 \leq y_2$ . Consider the two statements: (I)  $R$  is reflexive but not symmetric. (II)  $R$  is transitive. Then which one of the following is true?

- (1) Both (I) and (II) are correct. (2) Only (II) is correct.  
(3) Neither (I) nor (II) is correct. (4) Only (I) is correct.

**Q71.** Let  $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$  and  $B = I + \text{adj}(A) + (\text{adj } A)^2 + \dots + (\text{adj } A)^{10}$ . Then, the sum of all the elements of the matrix  $B$  is:

- (1) -124 (2) 22  
(3) -88 (4) -110

**Q72.** Given that the inverse trigonometric function assumes principal values only. Let  $x, y$  be any two real numbers in  $[-1, 1]$  such that  $\cos^{-1} x - \sin^{-1} y = \alpha$ ,  $-\frac{\pi}{2} \leq \alpha \leq \pi$ .

Then, the minimum value of  $x^2 + y^2 + 2xy \sin \alpha$  is

- (1) 0 (2) -1  
(3)  $\frac{1}{2}$  (4)  $-\frac{1}{2}$

**Q73.** If the function  $f(x) = \begin{cases} \frac{72^x - 9^x - 8^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}, & x \neq 0 \\ a \log_e 2 \log_e 3, & x = 0 \end{cases}$  is continuous at  $x = 0$ , then the value of  $a^2$  is equal to

- (1) 968 (2) 1152  
(3) 746 (4) 1250

**Q74.** Let  $f(x) = 3\sqrt{x-2} + \sqrt{4-x}$  be a real valued function. If  $\alpha$  and  $\beta$  are respectively the minimum and the maximum values of  $f$ , then  $\alpha^2 + 2\beta^2$  is equal to

- (1) 42 (2) 38  
(3) 24 (4) 44

**Q75.** If the value of the integral  $\int_{-1}^1 \frac{\cos \alpha x}{1+3^x} dx$  is  $\frac{2}{\pi}$ . Then, a value of  $\alpha$  is

- (1)  $\frac{\pi}{3}$  (2)  $\frac{\pi}{6}$   
(3)  $\frac{\pi}{4}$  (4)  $\frac{\pi}{2}$

**Q76.** The area (in sq. units) of the region described by  $\{(x, y) : y^2 \leq 2x, \text{ and } y \geq 4x - 1\}$  is

- (1)  $\frac{11}{32}$  (2)  $\frac{8}{9}$   
(3)  $\frac{11}{12}$  (4)  $\frac{9}{32}$

**Q77.** Let  $y = y(x)$  be the solution of the differential equation  $(x^2 + 4)^2 dy + (2x^3 y + 8xy - 2) dx = 0$ . If  $y(0) = 0$ , then  $y(2)$  is equal to

- (1)  $\frac{\pi}{32}$  (2)  $2\pi$   
(3)  $\frac{\pi}{8}$  (4)  $\frac{\pi}{16}$

**Q78.** Let  $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ ,  $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$  and  $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}$ ,  $x \in \mathbb{R}$ . If  $\vec{d}$  is the unit vector in the direction of  $\vec{b} + \vec{c}$  such that  $\vec{a} \cdot \vec{d} = 1$ , then  $(\vec{a} \times \vec{b}) \cdot \vec{c}$  is equal to

- (1) 11 (2) 3  
(3) 9 (4) 6

**Q79.** For  $\lambda > 0$ , let  $\theta$  be the angle between the vectors  $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$  and  $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$ . If the vectors  $\vec{a} + \vec{b}$  and  $\vec{a} - \vec{b}$  are mutually perpendicular, then the value of  $(14 \cos \theta)^2$  is equal to

- (1) 50 (2) 40  
(3) 25 (4) 20

**Q80.** Let P be the point of intersection of the lines  $\frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1}$  and  $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$ . Then, the shortest distance of P from the line  $4x = 2y = z$  is

- (1)  $\frac{5\sqrt{14}}{7}$  (2)  $\frac{3\sqrt{14}}{7}$   
 (3)  $\frac{\sqrt{14}}{7}$  (4)  $\frac{6\sqrt{14}}{7}$

**Q81.** There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is \_\_\_\_\_

**Q82.** Let  $S = \{\sin^2 2\theta : (\sin^4 \theta + \cos^4 \theta)x^2 + (\sin 2\theta)x + (\sin^6 \theta + \cos^6 \theta) = 0 \text{ has real roots}\}$ . If  $\alpha$  and  $\beta$  be the smallest and largest elements of the set  $S$ , respectively, then  $3((\alpha - 2)^2 + (\beta - 1)^2)$  equals \_\_\_\_\_

**Q83.** Consider a triangle ABC having the vertices  $A(1, 2)$ ,  $B(\alpha, \beta)$  and  $C(\gamma, \delta)$  and angles  $\angle ABC = \frac{\pi}{6}$  and  $\angle BAC = \frac{2\pi}{3}$ . If the points B and C lie on the line  $y = x + 4$ , then  $\alpha^2 + \gamma^2$  is equal to \_\_\_\_\_

**Q84.** Let A be a  $2 \times 2$  symmetric matrix such that  $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$  and the determinant of A be 1. If  $A^{-1} = \alpha A + \beta I$ , where I is an identity matrix of order  $2 \times 2$ , then  $\alpha + \beta$  equals \_\_\_\_\_

**Q85.** Consider the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \frac{2x}{\sqrt{1+9x^2}}$ . If the composition of

$$\underbrace{f, (f \circ f \circ f \circ \dots \circ f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}, \text{ then the value of } \sqrt{3\alpha + 1} \text{ is equal to } \underline{\hspace{2cm}}$$

**Q86.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a thrice differentiable function such that  $f(0) = 0$ ,  $f(1) = 1$ ,  $f(2) = -1$ ,  $f(3) = 2$  and  $f(4) = -2$ . Then, the minimum number of zeros of  $(3f'f'' + ff''')(x)$  is \_\_\_\_\_

**Q87.** If  $\int \operatorname{cosec}^5 x dx = \alpha \cot x \operatorname{cosec} x \left(\operatorname{cosec}^2 x + \frac{3}{2}\right) + \beta \log_e \left|\tan \frac{x}{2}\right| + C$  where  $\alpha, \beta \in \mathbb{R}$  and C is the constant of integration, then the value of  $8(\alpha + \beta)$  equals \_\_\_\_\_

**Q88.** Let  $y = y(x)$  be the solution of the differential equation  $(x + y + 2)^2 dx = dy$ ,  $y(0) = -2$ . Let the maximum and minimum values of the function  $y = y(x)$  in  $[0, \frac{\pi}{3}]$  be  $\alpha$  and  $\beta$ , respectively. If  $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$ ,  $\gamma, \delta \in \mathbb{Z}$ , then  $\gamma + \delta$  equals \_\_\_\_\_

**Q89.** Consider a line L passing through the points  $P(1, 2, 1)$  and  $Q(2, 1, -1)$ . If the mirror image of the point  $A(2, 2, 2)$  in the line L is  $(\alpha, \beta, \gamma)$ , then  $\alpha + \beta + 6\gamma$  is equal to \_\_\_\_\_

**Q90.** In a tournament, a team plays 10 matches with probabilities of winning and losing each match as  $\frac{1}{3}$  and  $\frac{2}{3}$  respectively. Let  $x$  be the number of matches that the team wins, and  $y$  be the number of matches that team loses. If the probability  $P(|x - y| \leq 2)$  is  $p$ , then  $3^9 p$  equals \_\_\_\_\_